Solution to HW IV

To formally prove that the collection of typed graphs over a fixed type graph TG, together with their type-preserving morphisms, forms a category, we must verify the following three fundamental properties.

1. Existence of Identity Morphisms

For any typed graph (G, type), the identity morphism is defined as:

$$\mathrm{id}_G = (\mathrm{id}_V, \mathrm{id}_E)$$

where:

- $\mathrm{id}_V(v) = v$ for all $v \in V$
- $\mathrm{id}_E(e) = e$ for all $e \in E$

This identity morphism trivially preserves the graph structure:

$$s \circ \mathrm{id}_E = \mathrm{id}_V \circ s = s$$

$$t \circ \mathrm{id}_E = \mathrm{id}_V \circ t = t$$

It also preserves the types for both vertices and edges, satisfying the commutativity requirement:

$$\mathrm{type}_V \circ \mathrm{id}_V = \mathrm{type}_V$$

$$\mathrm{type}_E \circ \mathrm{id}_E = \mathrm{type}_E$$

Hence, id_G is a valid typed graph morphism.

2. Closure under Composition

Given two valid morphisms of typed graphs:

$$f : (G_1, \mathrm{type}_1) \to (G_2, \mathrm{type}_2) \quad \text{and} \quad g : (G_2, \mathrm{type}_2) \to (G_3, \mathrm{type}_3)$$

We define their composition as:

$$g \circ f = (g_V \circ f_V, g_E \circ f_E)$$

First, we verify the preservation of the graph structure:

$$s_3 \circ (g_E \circ f_E) = g_V \circ f_V \circ s_1$$

$$t_3 \circ (g_E \circ f_E) = g_V \circ f_V \circ t_1$$

Next, we verify the preservation of types for both vertices and edges by chaining the equalities from both morphisms:

$$\text{type}_{3,V} \circ (g_V \circ f_V) = \text{type}_{2,V} \circ f_V = \text{type}_{1,V}$$

$$\text{type}_{3,E} \circ (g_E \circ f_E) = \text{type}_{2,E} \circ f_E = \text{type}_{1,E}$$

Therefore, the composition $g \circ f$ is a valid morphism of typed graphs.

3. Associativity

Composition in this category is associative because the underlying function composition is inherently associative:

$$h \circ (g \circ f) = (h \circ g) \circ f$$

for any consecutive sequence of morphisms:

$$f : G_1 \to G_2, \quad g : G_2 \to G_3, \quad h : G_3 \to G_4$$

Since all three axioms hold, we conclude that typed graphs over TG form a valid category.